FOOD DELIVERY PERFORMANCE ANALYTICS

Power BI Project Report

Student Name	Charanjot Singh Walia
Registration Number	PIET24CD019
Project Title	Food Delivery Performance Analytics

1. Project Overview

This project presents an interactive Power BI dashboard for analysing food delivery operations. It combines headline KPIs with visual breakdowns of order status, meal periods, delivery partners, monthly revenue, weekday activity and delivery-time categories. Month and Payment Mode slicers allow users to explore results for selected periods and payment channels.

2. Dataset Description

The dashboard indicates a food-delivery dataset containing order-level information and related customer, menu, payment, time and delivery-partner attributes. Fields needed to support the visuals likely include Order ID, Customer ID, Order Date/Month, Revenue or Order Amount, Order Status, Meal Period, Payment Mode, Delivery Partner, Delivery Time and Menu Item. The source file, exact row count, column names and date range were not supplied; this description is based on the dashboard.

3. Project Objectives

• Monitor customers, orders, revenue, menu items and average delivery time.

• Understand delivered, cancelled, pending and returned order proportions.

• Compare order distribution across morning, afternoon, evening and night.

• Evaluate order contribution by delivery partner and payment mode.

• Track monthly revenue and compare activity across weekdays.

• Analyse delivery-time buckets to identify possible fulfilment delays.

4. Data Cleaning and Preparation

The screenshot does not expose Power Query steps. The following is a recommended checklist to verify against the PBIX: remove duplicate order records using the correct order key; handle blank or invalid customer, date, amount and delivery-time values; assign correct data types; standardise status, payment-mode and partner labels; check negative or impossible amounts/times; derive Month, Day Name and Meal Period where required; and reconcile totals with the source. Keep only steps that were actually performed in the final submission.

5. Data Model

A suitable model is a star schema with an Orders fact table at one row per order and supporting Customer, Date, Menu Item, Delivery Partner and Payment Mode dimensions. The fact table stores transactional values such as revenue, delivery time and status. Each dimension should have a unique key and a one-to-many relationship to Orders. This is a proposed model; confirm actual table names, keys and relationships in Power BI Model view.

Table	Purpose / example fields
Fact_Orders	OrderID, CustomerID, DateKey, MenuItemID, PartnerID, PaymentModeID, Revenue, Status, DeliveryTime
Dim_Customer	CustomerID and customer attributes
Dim_Date	DateKey, Date, Month, Month Number, Day Name, Day Number
Dim_Menu	MenuItemID and menu item attributes
Dim_DeliveryPartner	PartnerID and delivery partner name
Dim_PaymentMode	PaymentModeID and payment mode

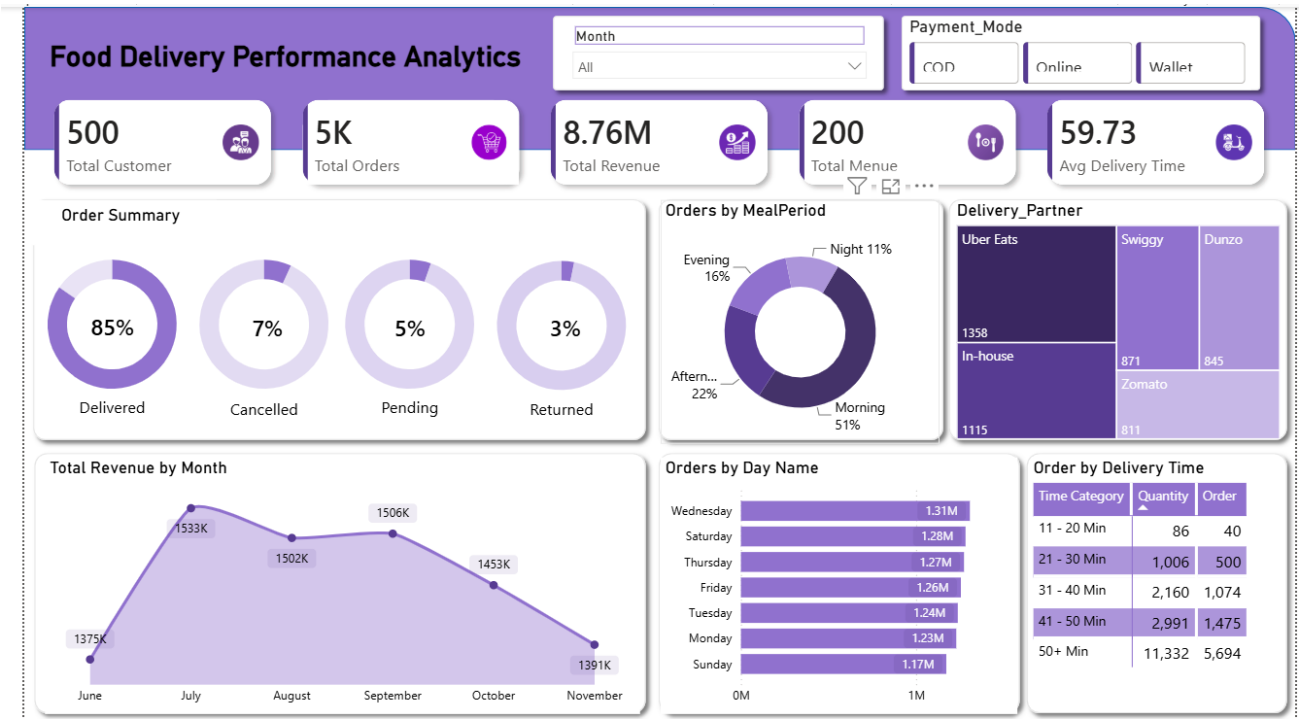
6. DAX Measures

These are representative measures for the displayed KPIs. Replace table and column names with the exact names used in the project; they are not extracted from the PBIX.

Measure	Example DAX
Total Customers	Total Customers = DISTINCTCOUNT(Orders[CustomerID])
Total Orders	Total Orders = DISTINCTCOUNT(Orders[OrderID])
Total Revenue	Total Revenue = SUM(Orders[Revenue])
Total Menu Items	Total Menu Items = DISTINCTCOUNT(Menu[MenuItemID])
Avg Delivery Time	Avg Delivery Time = AVERAGE(Orders[DeliveryTime])
Delivered %	Delivered % = DIVIDE(CALCULATE([Total Orders], Orders[Status] = "Delivered"), [Total Orders], 0)
Cancelled %	Cancelled % = DIVIDE(CALCULATE([Total Orders], Orders[Status] = "Cancelled"), [Total Orders], 0)
Pending %	Pending % = DIVIDE(CALCULATE([Total Orders], Orders[Status] = "Pending"), [Total Orders], 0)
Returned %	Returned % = DIVIDE(CALCULATE([Total Orders], Orders[Status] = "Returned"), [Total Orders], 0)

7. Dashboard Screenshot

The supplied dashboard contains KPI cards, order-status doughnut charts, meal-period distribution, delivery-partner treemap, monthly revenue trend, weekday chart and delivery-time summary.



8. Key Findings

• The KPI cards show 500 total customers, approximately 5K orders, 8.76M total revenue, 200 menu items and an average delivery time of 59.73 (unit not specified).

• Delivered orders form 85% of the total; Cancelled is 7%, Pending 5% and Returned 3%.

• Morning is the largest meal-period segment at 51%, followed by Afternoon (22%), Evening (16%) and Night (11%).

• Uber Eats has the largest displayed partner count (1,358), followed by In-house (1,115), Swiggy (871), Dunzo (845) and Zomato (811).

• Among the displayed months, revenue peaks in July at about 1,533K and is lowest in June at about 1,375K. It remains relatively steady through September before falling in October and November.

• Wednesday has the highest displayed weekday value (1.31M), while Sunday is lowest (1.17M). The 50+ minute delivery bucket has the largest quantity (11,332), indicating a potential delivery-speed improvement area.

9. Conclusion

The dashboard provides a consolidated view of customer activity, order outcomes, revenue trends and delivery operations. The visible results show a high delivered-order share and strong morning activity, while the large 50+ minute delivery bucket highlights an area for operational review. Month and Payment Mode filters support interactive exploration and help stakeholders identify patterns and improvement opportunities.

Report note: Dataset schema, Power Query transformations, model relationships and original DAX expressions were not provided. Those sections are suggested documentation and should be checked against the original Power BI file.